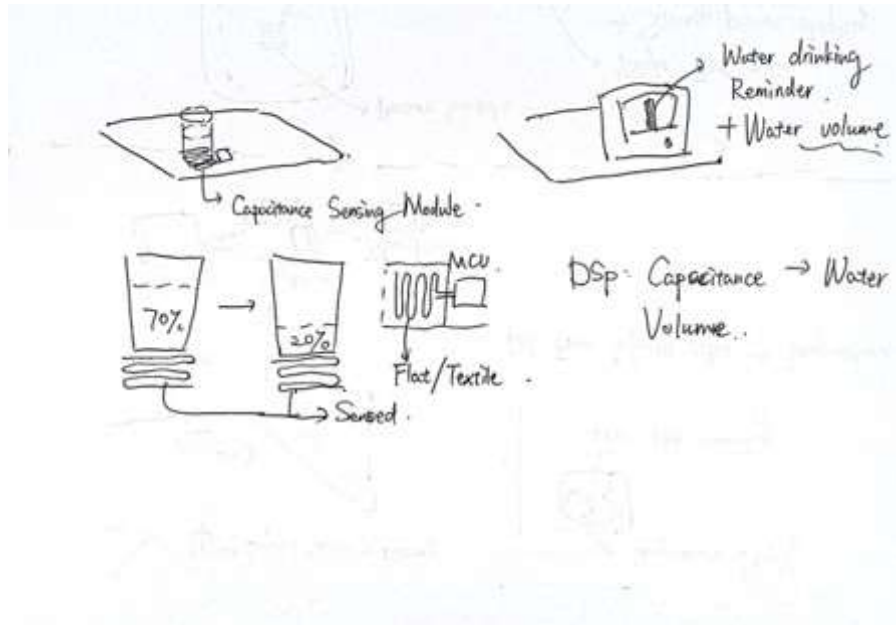


On-table drinking reminder

Overview

This project aims to design a wireless smart water monitoring system that tracks the water volume in a cup placed on a table and reminds users to drink water in a timely manner. By combining capacitive sensing, wireless communication, and algorithmic processing, the system supports healthier daily hydration habits with minimal user intervention.

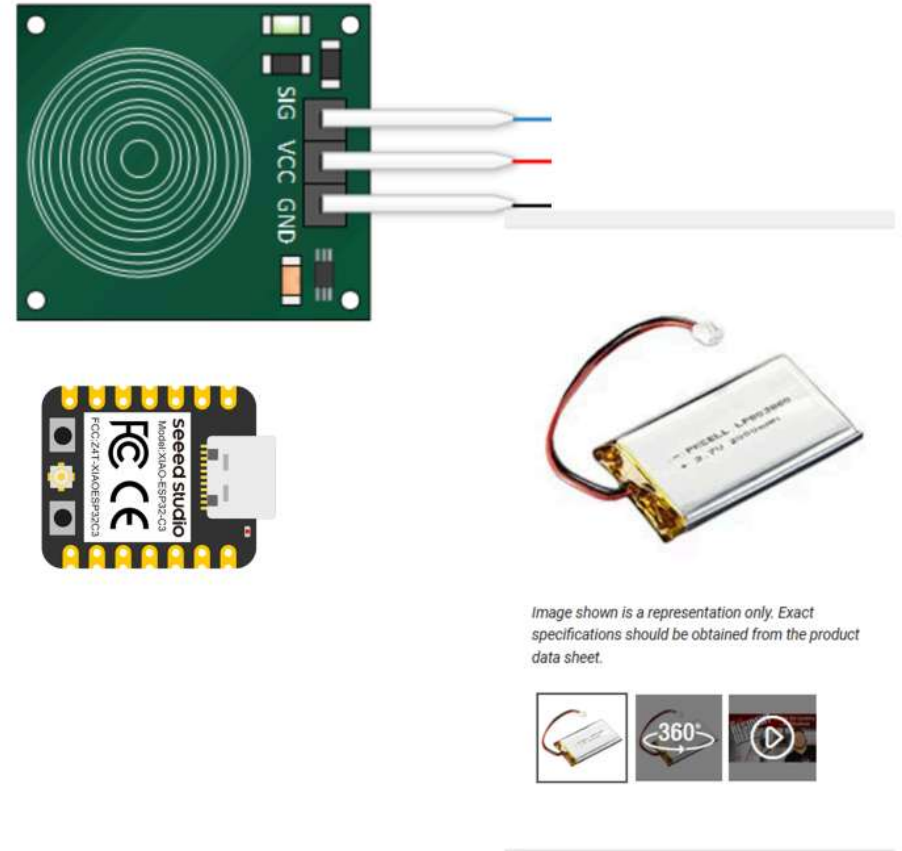


Idea

The core idea is to wirelessly monitor the water volume on a table (visualized by needle gauge) and provide intelligent drinking reminders based on real-time sensing and usage patterns. The system continuously estimates the remaining water volume and determines when a reminder (LED) should be triggered.

Sensor

- Capacitive sensor sense the capacitance
- Water has capacitance, different volume of water resulted in different capacitance, then the volume of the water can be detected by decoding the sensed capacitance by the sensor
- This module support decoding the capacitance via the transformed voltage, which can be sensed using Analog reading through MCU
- **Capacitive Sensor: TTP223**
- **MCU: Xiao ESP32C3**



Display

- Stepper-driven motor + Adafruit motor shield + LED + MCU
- The inner counter of MCU record the time between two times of drinking water and let the LED glow once the time interval is bigger than a threshold.
- The motor with gauge needle will rotate according to the water volume when water is stably placed on the capacitive sensor.
- **Motor: C140 motor**
- **LED: LSM0603 Series 0603 SMD LED**
- **Motor shield chip: TB6612FNG**
- **MCU: Xiao ESP32C3**

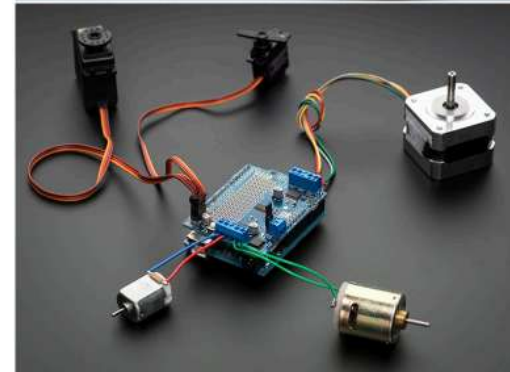
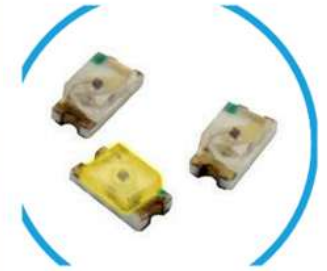


Image shown is a representation only. Exact specifications should be obtained from the product data sheet.



Figure & Diagram

